2011 F=ma Exam: Problem 11

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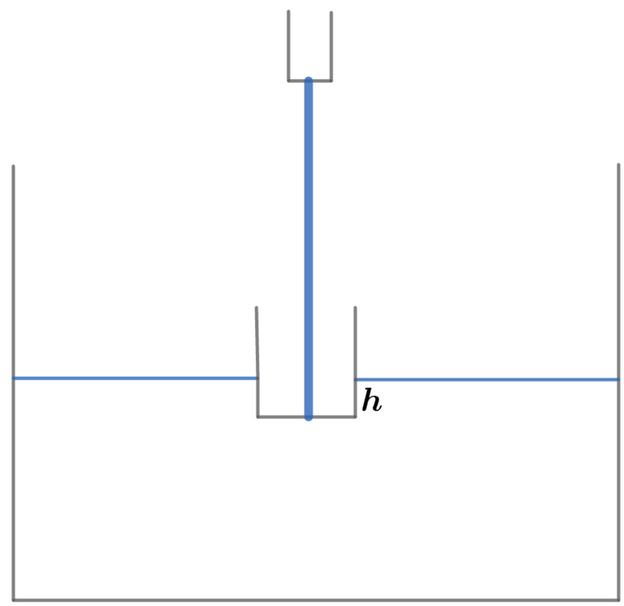
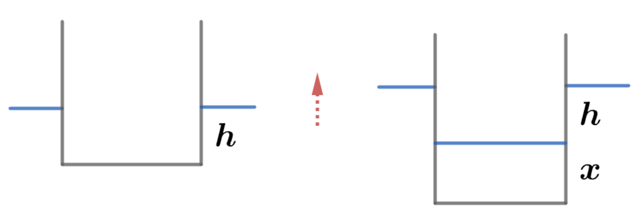
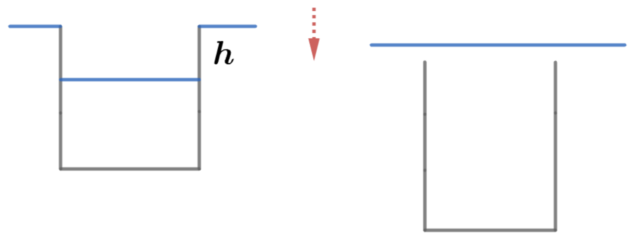
Initially, we have the metal cup floating in the tub with height difference h between the water level outside and inside the cup (the bottom of the cup before the tap is turned on).



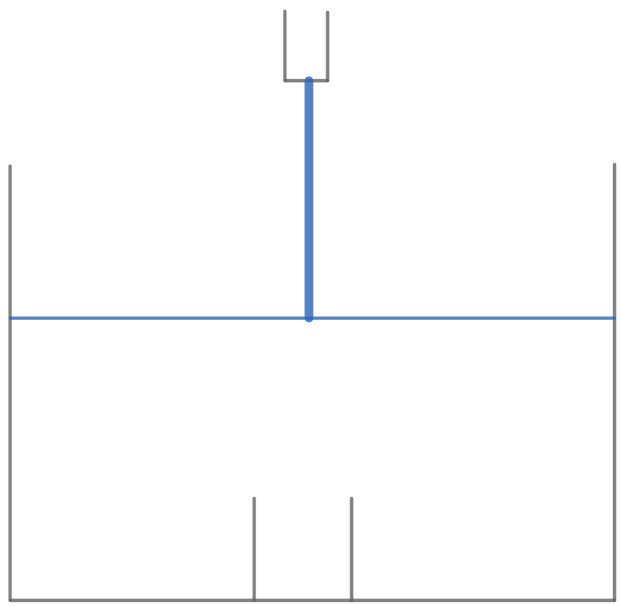
After the tap is turned on, the height difference between the water levels outside and inside stays constant at h. This is because when water of height x is added to the cup, the cup must also sink by x to displace the same weight of water and come back to equilibrium.



Since the cup sinks to displace the same amount of water that it contains, adding water to the cup changes the water level outside at the same rate as adding water directly to the tub. This continues until the critical moment when the top of the cup becomes level with the water outside.



At this moment, water is abruptly added the cup since it can flow in from the top in addition to the tap. The cup becomes submerged and the water level decreases a bit since the empty volume taken up by the cup across height h gets filled in.



Finally, the tap directly adds water to the tub. The water level rises at the same rate as when the tap added water to the cup. Thus, the water level over time is shown in graph $\boxed{\text{C}}$.